

Chapter 7

Spring Constant

A spring is an elastic object used to store mechanical energy. Springs are typically made of steel. In automobiles, springs in the suspension system help ensure that vibrations from the wheels are not directly transmitted to the vehicle body. In addition, springs also serve to improve tire grip on the road surface. The degree of extension or compression of a spring is expressed by a constant known as the spring constant (k). The value of k is highly important and must be taken into account whenever a spring is to be utilized.

7.1 Objectives

To be able to measure the spring constant.

7.2 Tools

1. Two springs of equal length.
2. Several weights.
3. Stand and holder.
4. Ruler.

7.3 Theoretical Foundation

According to Hooke's law, when a spring is subjected to a force F resulting in spring elongation (Δx), within its elastic limit, the elongation is always proportional to the applied force. Therefore, the following relationship holds:

$$F \propto \Delta x \quad (7.1)$$

If a proportional constant (k) is introduced into equation (7.1), then the equation becomes:

$$F = k\Delta x \quad (7.2)$$

The constant (k) represents the spring constant. Its value depends on the shape, size and material of the spring. If the applied force is the weight of a mass m , such that the presence of m causes an extension Δy , then the relationship can be written as:

$$mg = k\Delta y \quad (7.3)$$

Or

$$\Delta y = \left(\frac{g}{k}\right)m \quad (7.4)$$

7.4 Experimental Methods

7.4.1 Determination of a single spring constant

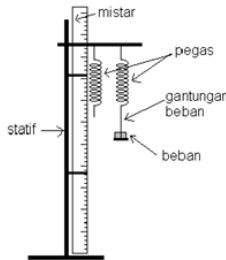


Figure 7.1: Schematic of the experimental set-up

1. Place a spring (as the first spring, its spring constant is denoted as k_1) in position and then record the scale at the end of the spring (y_0). Measure the unstretched length of that spring.
2. Then hang the weight holder and slotted weights, causing the spring will stretch and the end of the spring will now indicate a new scale reading y_1 .

3. The elongation of the spring is obtained from the difference between the two scale readings, that is $y_1 - y_0$. Use the maximum load mass that causes the elongation ($y_1 - y_0$) to be nearly equal to $2y_0$.
4. Perform step 3 by varying the mass from minimum to maximum, using approximately 10 different mass values.
5. Repeat steps 1 to 4 for the second spring, whose spring constant is denoted as k_2 .

7.4.2 Determining the spring constant for series and parallel arrangements

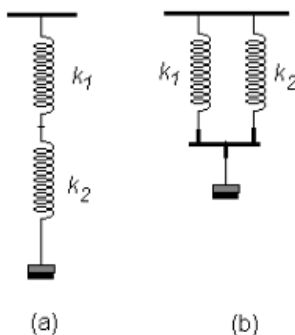


Figure 7.2: Compound spring arrangement

1. Arrange two springs in series as shown in figure 7.2a. Record the initial position of the end of the spring y_0 .
2. Then hang the weight holder and slotted mass. The end of the spring will point to a new scale reading y_1 . Record this reading. The extension is given by $\Delta y = y_1 - y_0$.
3. Add more weights and record the new scale reading indicated by the end of the spring.

4. Repeat the procedure for the parallel spring arrangement as shown in figure 7.2b.

Noted:

A spring may only be stretched up to a maximum of twice its original length. You must never exceed this limit, as overstretching will cause the spring to undergo deformation (it will not return to its original shape).

Data analysis

For the first and second springs, plot a graph of the spring elongation (Δy) as a function of the load mass (m). Then, from the slope of the graph, calculate the spring constant for each spring used (denoted as k_1 and k_2). In this experiment, the acceleration due to gravity at the location is known to be 980 cm/s^2 . Using the same method, plot spring length as a function of the load mass for the second step, namely, when k_1 and k_2 are arranged in series and in parallel (figure 7.2). From the slope of the Δy vs m graph, determine the spring constant for the arrangement. Once the spring constants for the series arrangement (denoted as k_s) and the parallel arrangement (denoted as k_p) are obtained, check whether the values of k_s and k_p correspond to the formulas that relate them to k_1 and k_2 :

$$k_s = \frac{k_1 k_2}{k_1 + k_2} \quad (7.5)$$

and

$$k_p = k_1 + k_2 \quad (7.6)$$

Draw the four graph lines corresponding to the values of k_1, k_2, k_s and k_p on a single graph.